Data Dictionary

MIMIC-III Clinical Database (v1.4)
Bayesian Hierarchical Survival Modeling

PHP2530 Final Project

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1 Overview and Data Structure

This data dictionary documents all variables extracted from the MIMIC-III Clinical Database for Bayesian hierarchical survival modeling. The preprocessing script (`data-prep.R`) creates three output datasets with distinct mathematical structures:

1.1 Conceptual Framework

For patient $i = 1, \ldots, N$ with T_i discrete time periods, we define:

1. **Static Baseline Covariate Matrix $\mathbf{X}$** (`ready_df`)
   - Dimension: $N \times P$ where N = number of admissions, P = number of covariates
   - Each row represents ONE admission at "Time Zero" (ICU admission)
   - Contains: demographics, aggregated vitals/labs, comorbidities, procedures, interventions
   - These covariates are **constant within patient** — they do not change over time
   - In person-period format, X_i is **repeated across all periods** for patient i

2. **Time-Varying Covariate Matrix $\mathbf{Z}_{i,t}$** (`longitudinal_df`)
   - Structure: One row per measurement occasion (patient $\times$ time point)
   - $Z_{i,t}$ = covariate vector for patient i at time t
   - Contains: vital signs (HR, BP, SpO2, temp, RR) with precise timestamps
   - These covariates **change over time** — different values at each t
   - Missing data is common (not all vitals measured at every time point)

3. **Discrete-Time Survival Format** (`person_period_df`)
   - Dimension: $\sum_{i=1}^{N} T_i$ rows (one row per patient per period at risk)
   - Structure: (i, t) indexed — patient i, period t
   - Contains: discrete outcome $y_{i,t}$, static covariates X_i, period-aggregated $Z_{i,t}$
   - Designed for Polya-Gamma augmented Gibbs sampling

1.2 Data Source Options

The preprocessing script supports two data sources via `DATA_SOURCE`:
- `DATA_SOURCE <- "demo"`: MIMIC-III Demo (∼100 patients) — for development
- `DATA_SOURCE <- "full"`: Complete MIMIC-III from PhysioNet (∼46,000 patients)

1.3 Output Files

All files saved to `data/` as R serialized objects:
- `ready_df.rds`: Static baseline matrix $\mathbf{X}$
- `longitudinal_df.rds`: Time-varying measurements $\mathbf{Z}_{i,t}$
- `person_period_df.rds`: Discrete-time survival format

2 Time Zero and Survival Framework

2.1 Time Zero Definition

Time Zero is defined as the **first ICU admission** within each hospital stay (`icu_intime`). This establishes a coherent survival framework:

Component	Definition
Time origin	First ICU admission timestamp (`icu_intime`)
Event	Death during hospitalization (`event_status = 1`)
Censoring	Discharge alive (`event_status = 0`)
Time scale	Hours from ICU admission to death/discharge

Justification: The ICU admission marks entry to highest-acuity care. Survival from this point is the clinically relevant outcome. Using the *first* ICU stay avoids survivorship bias from conditioning on patients who survived long enough for subsequent transfers.

2.2 Continuous-Time Outcome

For the continuous-time framework (`ready_df`):
- `time_to_event_hours`: T_i = hours from ICU admission to event/censoring
- `event_status`: $\delta_i = 1$ if died, $\delta_i = 0$ if censored

2.3 Discrete-Time Outcome

For the discrete-time framework (`person_period_df`), continuous time is discretized into 24-hour periods:
- Period 1: hours 0–24, Period 2: hours 24–48, etc.
- Maximum 7 periods (capped for computational tractability)
- `y_it`: binary outcome — 1 if event in period t, 0 otherwise

Example:

Patient Type	$y_{i,1}$	$y_{i,2}$	$y_{i,3}$	$y_{i,4}$	Interpretation
Dies in period 3	0	0	1	—	Event in final period
Censored in period 4	0	0	0	0	No event (discharged alive)

3 Static Baseline Matrix X (ready_df)

This dataset contains **one row per hospital admission**. All variables represent information available at or aggregated across the ICU stay, treated as "baseline" for modeling purposes.

3.1 Identification Variables

Variable	Type	Description
subject_id	Factor	Unique patient identifier (for patient-level random effects)
hadm_id	Factor	Unique hospital admission identifier (unit of analysis)
icustay_id	Integer	First ICU stay identifier within admission

3.2 Demographics

Variable	Type	Description
age	Numeric	Age at admission (years), capped at 90 per MIMIC convention
scaled_age	Numeric	Standardized age (mean=0, SD=1)
gender	Factor	Patient sex (M/F)
ethnicity	Factor	Self-reported ethnicity/race
marital_status	Factor	Marital status at admission
insurance	Factor	Primary insurance type

3.3 Admission & ICU Variables

Variable	Type	Description
admission_type	Factor	ELECTIVE, EMERGENCY, URGENT
admittime	POSIXct	Hospital admission timestamp
dischtime	POSIXct	Hospital discharge timestamp
deathtime	POSIXct	Time of death (NA if survived)
icu_intime	POSIXct	First ICU admission (Time Zero)
icu_outtime	POSIXct	First ICU discharge
first_careunit	Factor	ICU type: MICU, SICU, CCU, CSRU, TSICU
first_service	Factor	First hospital service
los_icu	Numeric	First ICU length of stay (days)
scaled_los_icu	Numeric	Standardized ICU LOS
n_icu_stays	Integer	Total ICU stays during admission
total_icu_los	Numeric	Total ICU days across all stays
is_surgical	Binary	1 if any surgical service involvement
n_service_changes	Integer	Number of service transfers
n_transfers	Integer	Number of unit transfers
n_unique_units	Integer	Distinct care units visited

3.4 Survival Outcomes

Variable	Type	Description
time_to_event_hours	Numeric	Hours from ICU admission to death/discharge
time_to_event_days	Numeric	Days from ICU admission to death/discharge
event_status	Binary	1 = died ($\delta_i = 1$), 0 = censored
icu_mortality	Binary	1 = died during ICU stay (stricter definition)
hospital_expire_flag	Binary	1 = died during hospitalization
end_time	POSIXct	Event/censoring timestamp

3.5 Vital Sign Summaries

Per-admission aggregates from CHARTEVENTS. Naming pattern: `vital_statistic`.

`hr_mean_val, hr_min_val, hr_max_val, hr_sd_val, hr_n_obs`

Vital Sign	CareVue ID	MetaVision ID	Units
hr (heart rate)	211	220045	bpm
sbp	51	220050	mmHg
dbp	8364	220051	mmHg
map	52	220052	mmHg
resp_rate	618	220210	breaths/min
spo2	646	220277	%
temp_c	676, 678→C	223762	°C
gcs	198	220739	3–15

Statistics computed:

- `mean_val`: Average across all measurements
- `min_val, max_val`: Extremes (capture decompensation)
- `sd_val`: Variability (instability indicator)
- `n_obs`: Number of observations (data completeness)

3.6 Laboratory Summaries

Per-admission aggregates from LABEVENTS. Naming pattern: `lab_statistic`.

```
lactate_first_val, lactate_last_val, lactate_min_val, lactate_max_val,
lactate_mean_val, lactate_n_labs
```

Lab Test	ID	Units	Clinical Relevance
lactate	50813	mmol/L	Tissue perfusion, sepsis marker
creatinine	50912	mg/dL	Renal function (SOFA)
bilirubin	50885	mg/dL	Hepatic function (SOFA)
platelets	51265	K/uL	Coagulation (SOFA)
wbc	51301	K/uL	Infection/immune
hemoglobin	51222	g/dL	Anemia, bleeding
glucose	50931	mg/dL	Metabolic status
bun	51006	mg/dL	Renal function
sodium	50983	mEq/L	Electrolyte balance
potassium	50971	mEq/L	Cardiac/renal
pao2	50821	mmHg	Oxygenation (SOFA resp)
pco2	50818	mmHg	Ventilation
ph	50820	—	Acid-base status

Statistics computed:

- `first_val`: Baseline at admission
- `last_val`: Final status
- `min_val, max_val, mean_val, n_labs`: As above

3.7 Comorbidity Flags (ICD-9)

Variable	Type	ICD-9 Pattern	Description
n_diagnoses	Integer	—	Total diagnosis codes
icd_sepsis	Binary	995.91, 995.92, 038.x	Sepsis/septicemia
icd_heart_failure	Binary	428.x	CHF
icd_aki	Binary	584.x	Acute kidney injury
icd_ckd	Binary	585.x	Chronic kidney disease
icd_resp_failure	Binary	518.8x	Respiratory failure
icd_pneumonia	Binary	480–486	Pneumonia
icd_diabetes	Binary	250.x	Diabetes mellitus
icd_liver_disease	Binary	571.x	Chronic liver disease
icd_malignancy	Binary	140–208	Cancer
icd_stroke	Binary	430–438	Cerebrovascular disease
icd_mi	Binary	410.x	Myocardial infarction
icd_copd	Binary	490–496	COPD
comorbidity_count	Integer	—	Sum of ICD flags (0–12)
scaled_comorbidity	Numeric	—	Standardized burden

3.8 Procedures & Interventions

Variable	Type	Description
n_procedures	Integer	Total ICD-9 procedure codes
is_ventilated	Binary	Mechanical ventilation (96.7x)
received_dialysis	Binary	Hemodialysis (39.95)
has_central_line	Binary	Central venous catheter
received_transfusion	Binary	Blood transfusion (99.0x)
received_vasopressors	Binary	Any vasopressor administered
n_vasopressor_doses	Integer	Total vasopressor events
vasopressor_duration_hrs	Numeric	Total hours on vasopressors

3.9 DRG Severity Scores

Variable	Type	Description
drg_mortality_score	Integer	DRG mortality risk (1–4, higher = worse)
drg_severity_score	Integer	DRG severity (1–4, higher = more severe)

4 Time-Varying Matrix Z (longitudinal_df)

This dataset contains **one row per vital sign measurement occasion**. Structure: $Z_{i,t}$ = covariate vector for patient i at time t.

4.1 Key Variables

Variable	Type	Description
subject_id	Factor	Patient identifier
hadm_id	Factor	Admission identifier
icustay_id	Integer	ICU stay identifier
charttime	POSIXct	Measurement timestamp
hours_since_icu_admit	Numeric	t = hours since ICU admission (time index)
scaled_time	Numeric	Standardized time

4.2 Vital Signs (Raw Values)

Variable	Type	Description
hr	Numeric	Heart rate (bpm)
sbp	Numeric	Systolic blood pressure (mmHg)
dbp	Numeric	Diastolic blood pressure (mmHg)
map	Numeric	Mean arterial pressure (mmHg)
resp_rate	Numeric	Respiratory rate (breaths/min)
spo2	Numeric	Oxygen saturation (%)
temp_c	Numeric	Temperature (°C)
gcs	Numeric	Glasgow Coma Scale (3–15)

4.3 Scaled Vitals

`scaled_hr`, `scaled_sbp`, `scaled_dbp`, `scaled_map`, `scaled_resp_rate`, `scaled_spo2`, `scaled_temp_c`

4.4 Outcome Information (Merged from Cohort)

Variable	Type	Description
event_status	Binary	1 = died, 0 = censored
time_to_event_hours	Numeric	Patient's total survival time
end_time	POSIXct	Death or discharge timestamp
first_careunit	Factor	ICU type
age	Numeric	Age at admission
gender	Factor	Patient sex

4.5 Filtering Applied

The dataset is restricted to:

- Measurements after ICU admission (`hours_since_icu_admit` ≥ 0)
- First 7 days post-ICU admission (`hours_since_icu_admit` ≤ 168)

5 Discrete-Time Format (person_period_df)

This dataset contains **one row per patient per discrete time period**. Designed for discrete-time survival modeling with Polya-Gamma augmentation.

5.1 Why Discrete-Time Survival?

Continuous-time survival models (Cox, Weibull) can be computationally intensive in Bayesian settings. Discrete-time models offer advantages:
1. **Logistic regression formulation**: Transforms survival into sequence of binary outcomes
2. **Polya-Gamma augmentation**: Enables conjugate Gibbs sampling (no Metropolis-Hastings)
3. **Natural for time-varying covariates**: One covariate value per discrete period
4. **Computational efficiency**: Essential for large datasets like MIMIC-III

5.2 Mathematical Formulation

The discrete-time hazard:

$$h_{it} = \Pr(T_i \in (a_{t-1}, a_t] \mid T_i > a_{t-1}, X_i, Z_{i,t})$$

Logistic regression model:

$$\text{logit}(h_{it}) = \alpha_t + X_i'\beta + Z_{i,t}'\gamma + u_i$$

where:
- α_t: period-specific baseline hazard (intercept)
- X_i: static covariates (repeated across periods)
- $Z_{i,t}$: time-varying covariates (different each period)
- $u_i \sim N(0, \sigma_u^2)$: patient-level random effect (frailty)

5.3 Polya-Gamma Augmentation

The logistic likelihood is not conjugate to Gaussian priors. Polya-Gamma augmentation (Polson et al., 2013) introduces auxiliary variables $\omega_{it} \sim \text{PG}(1, \eta_{it})$ that make the conditional posterior of β Gaussian:

$$\beta \mid \omega, y \sim \text{Normal}(\cdot, \cdot)$$

This enables **fully conjugate Gibbs sampling** — dramatically improving computational efficiency and mixing for large datasets.

5.4 Key Variables

Variable	Type	Description
hadm_id	Factor	Admission identifier
subject_id	Factor	Patient identifier
period	Integer	Time period: 1, 2, ..., 7 (days)
max_period	Integer	Total periods at risk for this patient
y_it	Binary	**Discrete outcome**: 1 if event in period t

Variable	Type	Description
event_status	Binary	Original continuous-time indicator

5.5 Static Covariates (from X)

These are the X_i variables **repeated across all periods** for the same patient:

Variable	Type	Description
age	Numeric	Age at admission
gender	Factor	Patient sex
first_careunit	Factor	ICU type
comorbidity_count	Integer	Number of comorbidities
received_vasopressors	Binary	Any vasopressor use
is_ventilated	Binary	Mechanical ventilation
icd_sepsis	Binary	Sepsis diagnosis
drg_mortality_score	Integer	DRG mortality risk
los_icu	Numeric	ICU length of stay

5.6 Time-Varying Covariates (from Z)

Period-level aggregations of vital signs — these are the $Z_{i,t}$ that **change across periods**:

Variable	Type	Description
hr_mean	Numeric	Mean heart rate in period t
hr_max	Numeric	Maximum heart rate in period t
hr_n	Integer	Number of HR measurements in period t
map_mean	Numeric	Mean MAP in period t
resp_rate_mean	Numeric	Mean respiratory rate in period t
spo2_mean	Numeric	Mean SpO2 in period t
temp_c_mean	Numeric	Mean temperature in period t

5.7 Missingness Indicators

Missing vitals in ICU settings may be informative (e.g., stable patients monitored less frequently). We create indicators:

Variable	Type	Description
hr_missing	Binary	1 if no HR measurements in period (before imputation)
temp_c_missing	Binary	1 if no temperature measurements in period
map_missing	Binary	1 if no MAP measurements in period

5.8 Missing Data Imputation

For time-varying covariates in person-period format:

1. **Create missingness indicators** (before imputation) — allows model to learn if missingness is prognostic
2. **LOCF (Last Observation Carried Forward)**: Within each patient, fill missing periods with most recent available value
3. **Population median**: For patients with no prior measurements, use population median

Note: LOCF artificially shrinks variance. Consider fully Bayesian imputation (treating missing values as latent parameters) for sensitivity analysis.

6 Discretized Variables for BART Modeling

This section documents the discretization of continuous variables into binary factor variables, following the pattern from `data_rebin.R`: `as.factor(ifelse(...))`.

6.1 Rationale for Binary Factorization

The approach uses simple binary indicators (0/1 factors) based on clinically meaningful thresholds:

1. **Clinical interpretability**: Binary thresholds match clinical decision points (e.g., "is tachycardic?" rather than continuous HR)
2. **SOFA/SIRS alignment**: Thresholds correspond to established scoring systems
3. **Simplicity**: Binary factors are efficient for BART tree splits
4. **Missing data handling**: NA converted to "unknown" factor level
5. **Consistency**: All variables become factors with clear 0/1 interpretation

Pattern used: `as.factor(ifelse(variable >= threshold, 1, 0))`

6.2 Variables Removed Before Factorization

The following variables were removed from `ready_df`:

Variable	Reason for Removal
admission_location	High-cardinality text (7+ categories)
discharge_location	High-cardinality text; **LEAKAGE**
language	High-cardinality character
religion	High-cardinality character
diagnosis	Free-text primary diagnosis
dob	POSIXct datetime — use derived factors
dod	POSIXct datetime — use `event_status`
expire_flag	Outcome indicator — **LEAKAGE**
has_chartevents_data	No variance (all ICU patients have data)

All original continuous variables (age, vitals, labs, counts) are also removed after creating binary factors.

6.3 Demographic Factors

Variable	Threshold	Clinical Meaning
age_elderly	age ≥ 65	Elderly patient (geriatric threshold)

Variable	Threshold	Clinical Meaning
age_very_old	age ≥ 80	Very old (increased frailty risk)

6.4 ICU Stay Factors

Variable	Threshold	Clinical Meaning
los_prolonged	LOS ≥ 3 days	Prolonged ICU stay
los_extended	LOS ≥ 7 days	Extended/chronic critical illness
multiple_icu_stays	n_icu_stays > 1	Multiple ICU admissions

6.5 Vital Sign Factors

All thresholds based on SIRS criteria and standard clinical cutoffs.

Variable	Threshold	Clinical Meaning
Glasgow Coma Scale		
gcs_impaired	GCS < 15	Any neurologic impairment
gcs_severe	GCS ≤ 8	Severe impairment (intubation threshold)
Heart Rate		
hr_tachy	HR > 100 bpm	Tachycardia
hr_brady	HR < 60 bpm	Bradycardia
hr_abnormal	HR < 60 or > 100	Any heart rate abnormality
Respiratory Rate		
rr_tachy	RR > 20 /min	Tachypnea (SIRS criterion)
rr_severe	RR > 30 /min	Severe tachypnea
Oxygen Saturation		
spo2_low	SpO2 $< 95\%$	Hypoxia
spo2_severe	SpO2 $< 90\%$	Severe hypoxia
Temperature		
temp_fever	Temp > 38°C	Fever (SIRS criterion)
temp_hypo	Temp < 36°C	Hypothermia (SIRS criterion)
temp_abnormal	Temp < 36 or > 38	Any temperature abnormality

6.6 Laboratory Factors

Thresholds based on SOFA criteria and standard reference ranges.

Variable	Threshold	Clinical Meaning
Renal Function		
bun_high	BUN > 20 mg/dL	Elevated BUN
bun_very_high	BUN > 40 mg/dL	Very high BUN
creat_elevated	Cr ≥ 1.2 mg/dL	Elevated creatinine (SOFA 1)

Variable	Threshold	Clinical Meaning
creat_dysfunction	Cr $\geq$ 2.0 mg/dL	Renal dysfunction (SOFA 2)
Glucose		
glucose_high	Glucose > 180 mg/dL	Hyperglycemia (ICU target exceeded)
glucose_low	Glucose < 70 mg/dL	Hypoglycemia
Hemoglobin		
hgb_anemia	Hgb < 10 g/dL	Anemia
hgb_severe_anemia	Hgb < 7 g/dL	Severe anemia (transfusion threshold)
Platelets		
plt_low	Plt < 150 K/uL	Thrombocytopenia (SOFA coag)
plt_severe	Plt < 50 K/uL	Severe thrombocytopenia
Electrolytes		
k_low	K^+ < 3.5 mEq/L	Hypokalemia
k_high	K^+ > 5.0 mEq/L	Hyperkalemia
na_low	Na^+ < 135 mEq/L	Hyponatremia
na_high	Na^+ > 145 mEq/L	Hypernatremia
White Blood Cells		
wbc_high	WBC > 12 K/uL	Leukocytosis (SIRS criterion)
wbc_low	WBC < 4 K/uL	Leukopenia (SIRS criterion)
wbc_abnormal	WBC < 4 or > 12	Any WBC abnormality

6.7 Variability and Complexity Factors

Variable	Threshold	Clinical Meaning
hr_unstable	HR SD > 15	Heart rate variability (instability)
temp_unstable	Temp SD > 0.5	Temperature variability
high_comorbidity	count $\geq$ 3	High comorbidity burden
very_high_comorbidity	count $\geq$ 5	Very high comorbidity
many_diagnoses	n_dx > 10	Many diagnoses
many_procedures	n_proc > 5	Many procedures
polypharmacy	n_rx > 20	Polypharmacy
drg_high_mortality	DRG mort $\geq$ 3	High DRG mortality risk
drg_high_severity	DRG sev $\geq$ 3	High DRG severity

6.8 Existing Binary Indicators (Converted to Factors)

The following variables were already 0/1 integers and are converted to factors:

- **ICD Comorbidities**: icd_sepsis, icd_heart_failure, icd_aki, icd_ckd, icd_resp_failure, icd_pneumonia, icd_diabetes, icd_liver_disease, icd_malignancy, icd_stroke, icd_mi, icd_copd
- **Interventions**: is_ventilated, received_dialysis, has_central_line, received_transfusion, received_vasopressors, is_surgical
- **Medications**: on_antibiotic, on_anticoagulant, on_insulin, on_sedative

- **Cultures**: positive_blood_culture, any_positive_culture
- **Other**: callout_requested

6.9 Categorical Variables (Multi-level Factors)

These remain as multi-level factors:

Variable	Levels
admission_type	ELECTIVE, EMERGENCY, URGENT
marital_status	SINGLE, MARRIED, DIVORCED, WIDOWED, etc.
ethnicity	WHITE, BLACK, HISPANIC, ASIAN, etc.
insurance	Medicare, Medicaid, Private, Government
first_careunit	MICU, SICU, CCU, CSRU, TSICU
first_service	MED, SURG, CMED, CSURG, etc.
gender	M, F

6.10 Missing Data Handling

For all factor variables, missing values are converted to an explicit "unknown" factor level using `fct_na_value_to_level()`.

7 Preprocessing Rationale

7.1 Age Capping

MIMIC-III shifts dates of birth for patients >89 years (privacy protection), resulting in apparent ages >200. We cap at 90:

```
age = ifelse(age > 150, 90, age)
```

Justification: Preserves information that patient is elderly without implausible values distorting regression models.

7.2 Vital Sign Harmonization

MIMIC-III has two charting systems (CareVue and MetaVision) with different item ID schemes. We map both to common variable names.

Temperature harmonization: CareVue recorded both Celsius and Fahrenheit. We convert Fahrenheit to Celsius: $(F - 32) \times 5/9$.

7.3 Variable Scaling

Continuous predictors are standardized (mean=0, SD=1) to facilitate weakly informative priors:

```
scaled_age = scale(age)[, 1]
```

Justification: With priors like $\beta \sim N(0, 2.5)$, standardization ensures consistent interpretation across all variables.

7.4 Missing Data Defaults

- **Binary indicators**: Default to 0 (absence of code = condition not present)
- **Count variables**: Default to 0
- **DRG scores**: Default to 1 (lowest severity) — conservative choice
- **Lab/vital summaries**: Remain NA — imputation strategy left to analysis phase

8 Hierarchical Structure

The data contains natural clustering that supports multi-level models:

1. **Patient level** (`subject_id`): Some patients have multiple admissions
2. **ICU level** (`first_careunit`): 5 unit types (MICU, SICU, CCU, CSRU, TSICU)
3. **Observation level**: For longitudinal_df and person_period_df, repeated measurements/periods

   Example hierarchical model:

$$h_{it} = h_0(t) \exp(X_i'\beta + Z_{i,t}'\gamma + u_i + v_{k[i]})$$

   where $u_i \sim N(0, \sigma_u^2)$ (patient frailty) and $v_k \sim N(0, \sigma_v^2)$ (ICU-level effect).

9 Modeling Recommendations

9.1 Suggested Workflow

1. **Exploratory**: Use `ready_df` for descriptive statistics, univariate survival curves
2. **BART modeling**: Use discretized categorical variables (`*_cat`) for clinically interpretable tree-based analysis
3. **Latent class discovery**: BART or Dirichlet process mixtures on `ready_df` to identify heterogeneous patient strata
4. **Discrete-time survival**: Bayesian logistic regression on `person_period_df` with Polya-Gamma augmentation
5. **Joint modeling**: Link vital trajectories (`longitudinal_df`) with survival via shared random effects

9.2 BART Implementation Notes

When fitting BART with the `BART` package in R:

1. **Exclude outcomes and IDs**: Remove `subject_id`, `hadm_id`, `icustay_id`, `icu_mortality`, `event_status`, `hospital_expire_flag`, and all time-to-event variables
2. **Exclude timestamps**: Remove all `POSIXct` columns (`admittime`, `dischtime`, `deathtime`, `icu_intime`, `icu_outtime`, `end_time`)
3. **All factors**: All predictor variables in `ready_df` are now factors (binary 0/1 or multi-level categorical)
4. **No continuous variables**: Continuous variables have been removed; use the binary factor versions
5. **Missing data**: The "unknown" factor level allows BART to learn from missingness patterns

   Example R code for BART preparation:

```
exclude <- c("subject_id", "hadm_id", "icustay_id",
             "icu_mortality", "event_status", "hospital_expire_flag",
```

```
              "time_to_event_hours", "time_to_event_days",
              "admittime", "dischtime", "deathtime",
              "icu_intime", "icu_outtime", "end_time")

y <- as.numeric(as.character(ready_df$icu_mortality))
X <- ready_df %>% select(-any_of(exclude))

# Fit logistic BART
bart_model <- lbart(x.train = X, y.train = y)
```

9.3 Prior Recommendations

For scaled continuous predictors:

$$\beta_j \sim \text{Normal}(0, 2.5)$$
$$\sigma_u \sim \text{Half-Cauchy}(0, 2.5)$$

For period-specific intercepts (baseline hazard):

$$\alpha_t \sim \text{Normal}(0, 2) \quad \text{or random walk prior}$$

10 Data Quality Notes

1. **NOTEEVENTS**: Empty in demo version (no clinical notes)
2. **Age >89**: Capped at 90 due to MIMIC de-identification
3. **ICD-9 codes**: Assigned retrospectively for billing; may not reflect status at ICU admission
4. **Dual charting**: CareVue and MetaVision merged via item ID mapping
5. **Lab missingness**: May be informative (stable patients get fewer labs)
6. **LOCF limitations**: Artificially reduces variance; use missingness indicators as covariates
7. **Demo vs. Full**: Demo has ∼100 patients; full has ∼46,000